a nonstationary environment, the expected value for a given state transition will change over time, which is depicted in this time series as a changing mean or baseline over time. The Q function is trying to learn the expected value for state-actions, and it can only converge if the state-action values are stationary, but in the multi-agent setting, the expected state-action values can change over time due to the evolving policies of other agents.

Figure 9.4. The Q function takes a state and produces state-action values (Q values), which are then used by the policy function to produce an action. Alternatively, we can directly train a policy function that operates on a state and returns a probability distribution over actions.

Figure 9.5. If each agent has an action space of size 4 (i.e., it is represented by a 4 element one-hot vector), the joint action space of two agents is 42 = 16, or 4N where N is the number of agents. This means the growth of the joint action space is exponential in the number of agents. The figure on the right shows the joint action space size for agents with individual action spaces of size 2. Even with just 25 agents, the joint action space becomes a 33,554,432 element one-hot vector, which is computationally impractical to work with.

Figure 9.6. In neighborhood MARL, each agent has a field of view (FOV) or neighborhood, and it can only see the actions of the other agents within this neighborhood. However, it may still get the full state information about the environment.

Figure 9.7. The neighborhood Q function for agent j accepts the current state and the joint-action vector for the other agents within its neighborhood (or field of view), denoted a–j. It produces Q values that get passed to the policy function that chooses the action to take.

Figure 9.8. Electrons are negatively charged elementary particles that surround the nucleus of every atom. They have a property called spin, and it can either be spin-up or spin-down. Since they are charged particles, they generate a magnetic field, and the direction they spin determines the orientation of the poles (north or south) of the magnetic field.